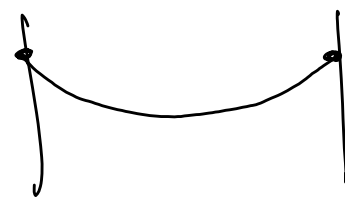
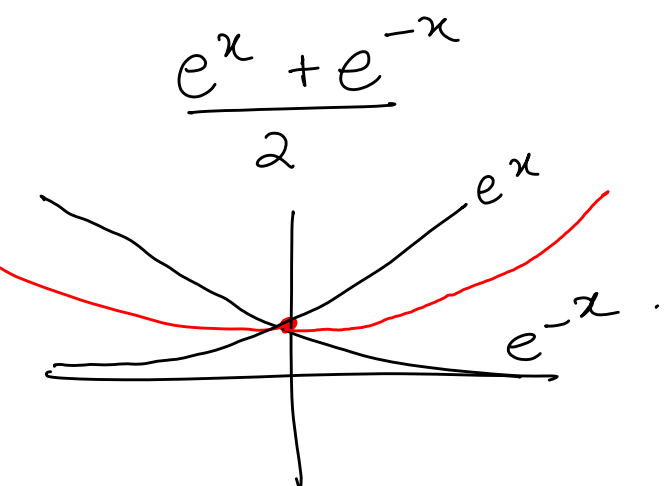


Unit 3.7 Hyperbolic functions.

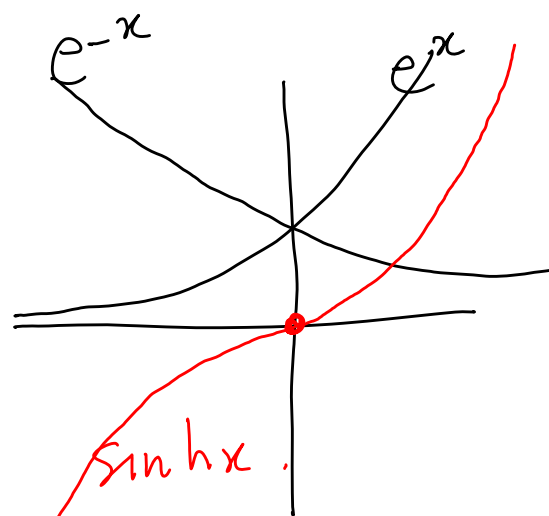
Note Title

2016/04/26



$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$



$$\begin{aligned} \frac{d}{dx} \cosh x &= \frac{d}{dx} \left(\frac{e^x + e^{-x}}{2} \right) \\ &= \frac{1}{2} (e^x - e^{-x}) = \sinh x. \end{aligned} \quad \begin{matrix} \text{(No)} \\ \text{(minus)} \end{matrix}$$

$$\begin{aligned} \frac{d}{dx} \sinh x &= \frac{d}{dx} \left(\frac{e^x - e^{-x}}{2} \right) \\ &= \frac{e^x + e^{-x}}{2} = \cosh x, \end{aligned}$$

Identities:

Recall: $\cos^2 x + \sin^2 x = 1$

Now: $\cosh^2 x - \sinh^2 x = 1.$

What about $\tanh x$?

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\Rightarrow \frac{d}{dx} \tanh x = \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x} = \frac{1}{\cosh^2 x}.$$

Differentiation exercises:

1. $f(x) = \ln(\cosh 3x)$

2. $f(x) = \sinh(\ln x) \cdot e^{\cosh x}$